

LUANG PHABANG, Lao People's Democratic Republic

WMO: 489300

Lat: 19.897N Lon: 102.161E Elev: 955 StdP: 14.20 Time Zone: 7.00 (E07) Period: 05-19 WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 51.5	(c) 55.0	(d) 46.3	(e) 48.0	(f) 63.7	(g) 49.5	(h) 54.3	(i) 62.3	(j) 12.7	(k) 56.8	(l) 9.6	(m) 68.9	(n) 1.5	(o) 140	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	18.3	100.1	76.9	97.8	76.4	95.4	76.4	81.8	91.4	80.7	90.5	79.9	89.1	5.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.4	157.8	85.7	78.5	153.2	84.4	77.8	149.6	83.4	46.5	92.0	45.3	90.6	44.5	89.7	90.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.7	8.8	7.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	79.1	69.8	74.9	78.8	83.8	84.4	83.7	82.9	82.4	82.4	80.2	75.3	70.6
	DBStd	6.23	4.99	3.80	5.26	3.79	3.63	2.75	2.65	2.23	2.17	2.79	4.33	4.30
	HDD50	1	1	0	0	0	0	0	0	0	0	0	0	0
	HDD65	32	19	1	3	0	0	0	0	0	0	0	0	9
	CDD50	10631	616	697	894	1014	1067	1011	1021	1004	973	935	759	640
	CDD65	5187	168	279	433	564	602	561	556	539	523	470	309	183
	CDH74	48092	927	2417	4254	6256	6653	5895	5449	4990	4797	3704	1910	840
	CDH80	20658	245	1114	2265	3386	3266	2480	2049	1755	1840	1464	619	175
(14) Wind	WSAvg	2.5	2.1	2.4	2.8	3.0	3.1	2.7	2.6	2.4	2.2	2.3	2.1	2.4
(15) Precipitation	PrecAvg	61.4	0.6	0.5	1.9	3.8	6.8	9.0	12.1	13.5	9.3	3.1	1.1	0.9
	PrecMax	75.1	2.8	3.0	6.1	7.2	11.1	20.1	16.7	21.1	19.0	8.1	4.6	5.9
	PrecMin	43.9	0.0	0.0	0.0	0.3	2.5	4.0	4.7	7.1	4.8	0.7	0.0	0.0
	PrecStd	8.8	0.8	0.7	1.5	1.8	2.2	3.2	3.0	3.7	3.0	1.9	1.1	1.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.4	96.0	99.3	103.7	102.4	97.2	97.0	95.0	94.6	93.6	91.4	87.7
		MCWB	70.7	72.0	72.8	76.6	79.1	79.2	77.9	79.5	79.5	77.8	76.0	72.8
	2%	DB	85.8	93.3	97.1	100.9	99.6	95.2	94.1	92.5	93.0	91.7	88.7	84.2
		MCWB	70.2	71.6	73.2	76.7	78.7	78.6	78.6	79.2	79.4	77.3	75.1	71.1
	5%	DB	82.9	90.7	95.1	98.5	96.7	93.2	91.8	90.8	91.2	89.9	86.2	82.0
		MCWB	69.3	70.8	73.2	76.3	78.4	78.6	79.0	79.1	78.8	76.7	73.9	69.9
	10%	DB	80.2	87.8	92.8	95.4	93.6	91.1	89.6	88.7	89.4	88.0	83.9	79.1
		MCWB	68.0	70.2	72.9	75.9	78.2	78.6	78.6	78.8	78.4	76.0	72.4	68.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.6	77.3	78.5	81.9	82.9	82.7	81.7	82.7	82.3	80.9	78.5	74.7
		MCDB	84.3	89.9	92.5	94.5	96.4	90.9	91.1	90.3	90.9	89.6	87.0	83.2
	2%	WB	72.1	74.7	76.4	79.6	80.8	80.8	80.4	80.9	80.6	79.2	76.8	72.9
		MCDB	80.9	87.9	90.1	93.6	92.5	90.4	89.3	89.0	89.6	88.1	85.0	81.1
	5%	WB	70.6	73.0	75.2	78.2	79.8	79.8	79.6	79.9	79.7	78.2	75.7	71.5
		MCDB	78.6	85.7	89.2	91.6	90.6	88.8	87.7	87.6	88.1	86.6	83.3	78.8
	10%	WB	69.2	71.4	74.0	77.2	79.0	79.0	78.9	79.0	78.9	77.2	74.5	70.2
		MCDB	76.7	83.1	87.6	89.7	89.1	87.5	86.4	86.2	86.7	84.9	81.1	76.9
(35) Mean Daily Temperature Range	5% DB	MDBR	21.7	28.4	25.7	22.9	18.3	15.1	13.8	13.8	14.9	17.4	19.2	18.9
		MCDBR	25.7	32.5	30.7	27.3	22.7	18.6	17.7	16.8	17.6	19.2	20.0	21.2
		MCWBR	11.2	12.7	10.5	8.1	6.2	5.4	5.1	5.8	5.9	7.0	7.7	9.2
	5% WB	MCDBR	21.9	29.2	28.2	24.6	19.6	16.5	15.0	15.4	16.4	18.0	18.3	19.2
		MCWBR	11.0	13.4	11.1	8.7	6.7	5.6	5.5	6.1	6.3	7.6	7.7	9.5
(40) Clear-Sky Solar Irradiance	taub		0.431	0.538	0.789	0.749	0.493	0.446	0.444	0.472	0.494	0.526	0.428	0.421
	taud		2.213	1.919	1.435	1.549	2.193	2.357	2.354	2.255	2.174	2.044	2.275	2.254
	Ebn at Noon		266	242	191	202	259	269	270	264	257	243	265	265
	Edh at Noon		42	60	101	91	47	40	40	44	48	52	40	40
(44) All-Sky Solar Radiation	RadAvg		1306	1561	1603	1728	1612	1410	1309	1347	1445	1399	1373	1229
(45)	RadStd		97	127	106	129	141	128	154	109	60	76	108	102

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	+0.49	N/A	N/A	+0.95	N/A	N/A	N/A	N/A	+277	+272